

# Data for Policy Analysis and Management

CLASS 3: MEASURES OF CENTRAL  
TENDENCY, VARIABILITY, AND  
DISTRIBUTIONS

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# Agenda

1. Why summarize data
2. Measures:
  - Central Tendency: mean, median, mode
  - Dispersion: SD, Range, IQR
  - Shape: distribution, kurtosis
3. Summary
4. Group Exercise
5. Lab Session: Measures

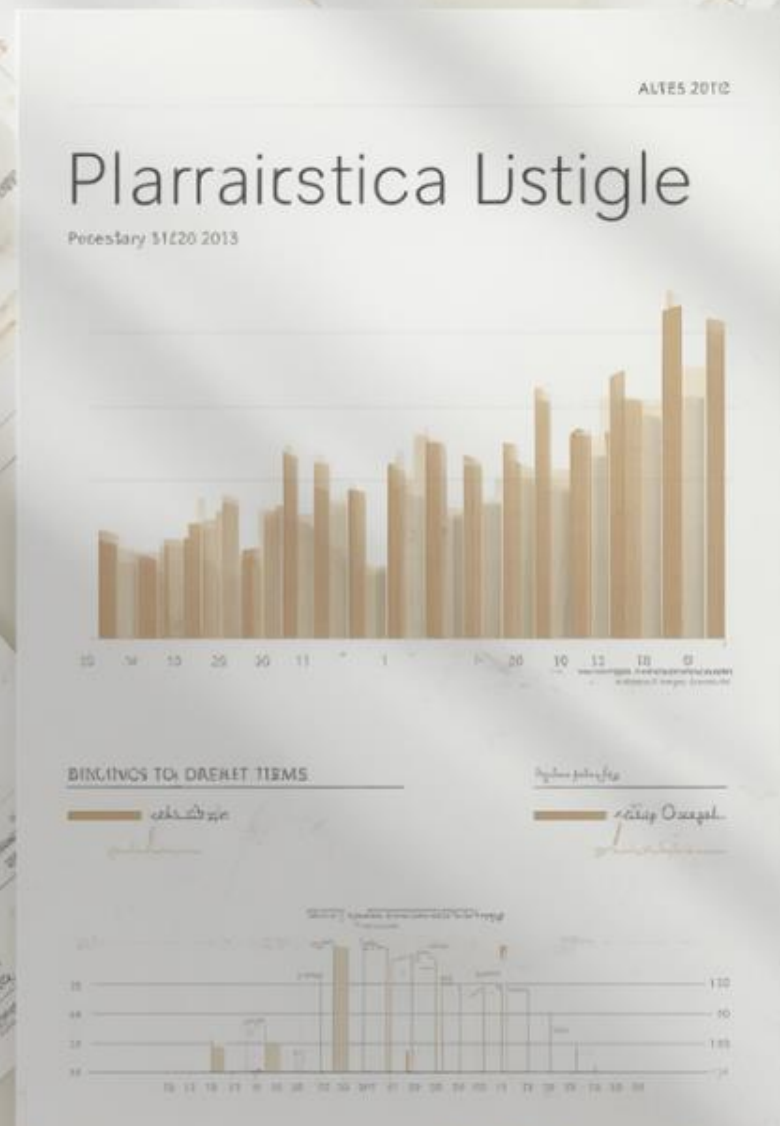
# Why We Summarize Data

## The Core Idea

- A variable is fully described by its distribution.
- A distribution shows: what values occur and how often.
- But full distributions can be too detailed to interpret directly.

## What We Do Instead

- Summarize distributions by shape, center, and spread.
- Reduce complexity while preserving key features.
  - Shape: reveals symmetry, skewness, and peaks
  - Center: locates the typical value
  - Spread: captures variability across observations



# Central Tendency: The Mean

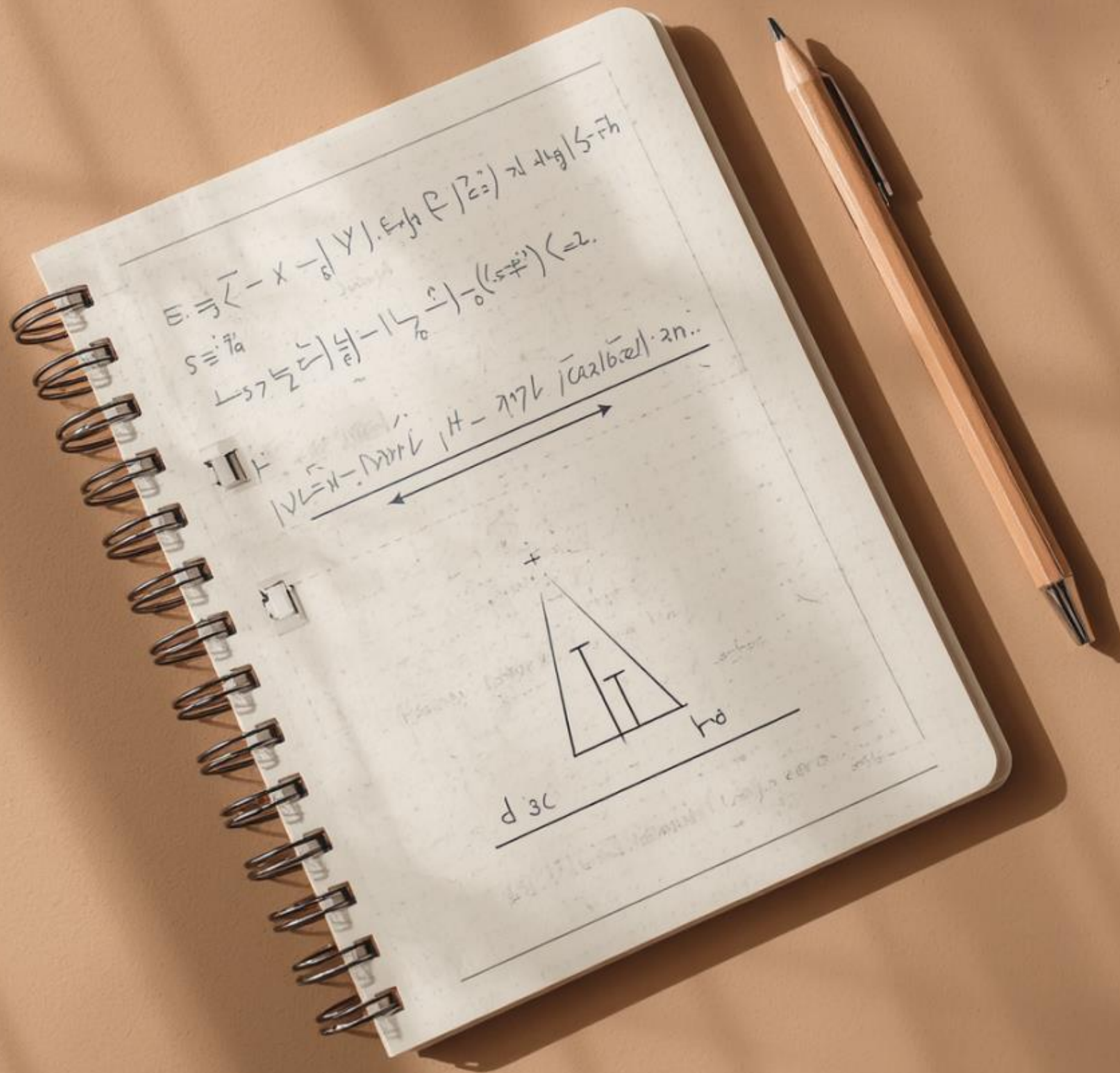
## Definition

- The mean is the sum of all values divided by the number of observations.
- It is interpreted as the balance point of a distribution (the point at which the distribution would be in equilibrium).

$$\bar{y} = \frac{\sum y_i}{n}$$

## Key Properties

- Uses all data points in its calculation.
- Sensitive to outliers and extreme values.
- Best suited for symmetric or roughly symmetric distributions.
- Not appropriate for nominal (categorical) data.
- In skewed distributions, the mean shifts toward the long tail.



# Central Tendency: The Median

## Definition

- The median is the middle value when data are ordered from smallest to largest.
- If
  - $n$  is odd: the middle observation.
  - $n$  is even: the average of the two middle values.

## Key Properties

- Robust to outliers: not pulled by extreme values
- Better than the mean for skewed distributions
- Works for ordinal and continuous data.
- Reflects the 50th percentile: half of observations fall below, half above.
- Rule of thumb:
  - skewed data → use median
  - symmetric data → use mean.

[illegible]

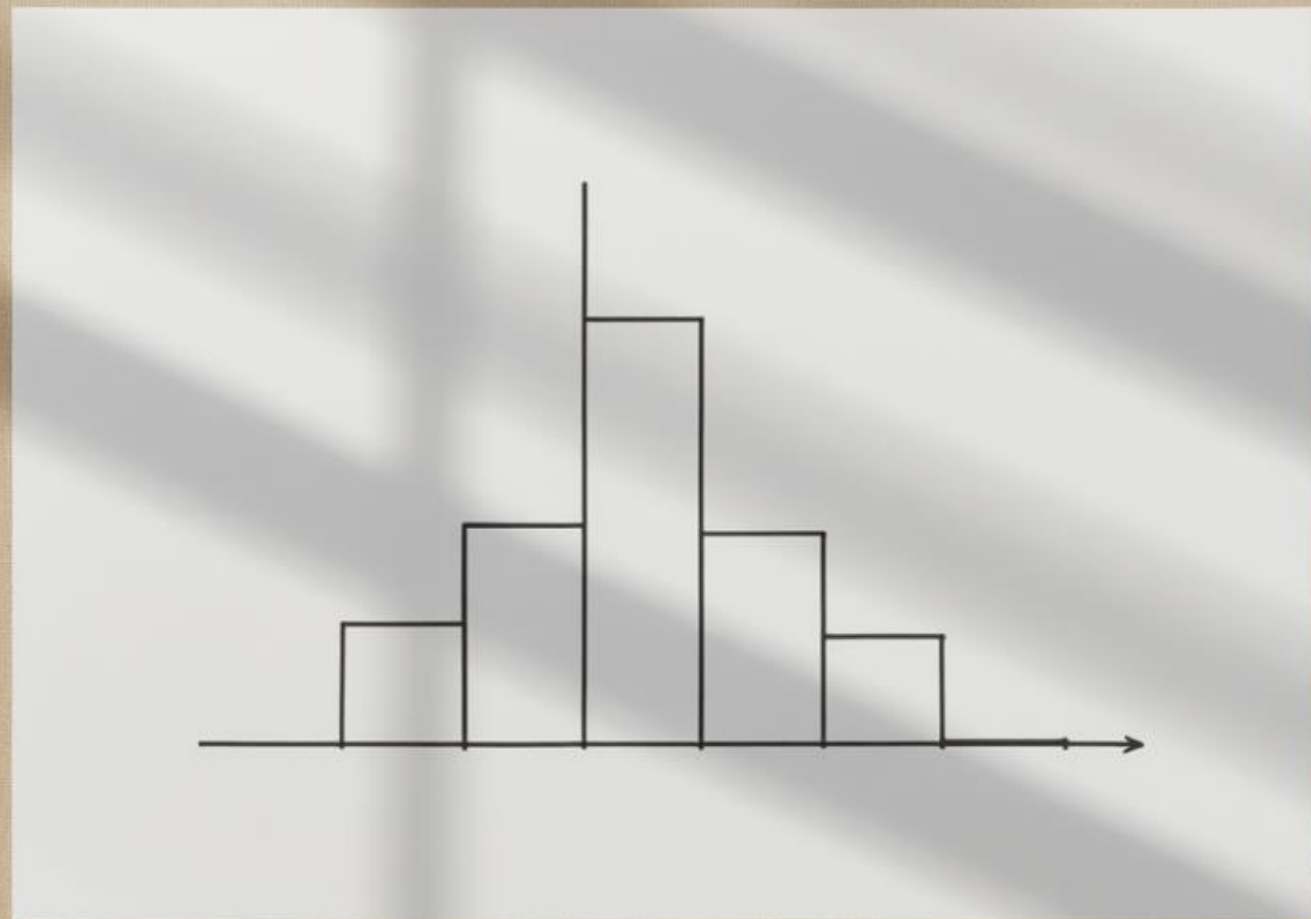
# Central Tendency: Mode - Most Frequent Value

## Definition

- The mode is the value that occurs most often.
- It is the only measure of central tendency that works for all data types: nominal, ordinal, and continuous.

## Key Properties

- Useful for categorical variables (e.g., most common childcare type, most common survey responses).
- Unimodal = one peak; bimodal = two peaks (suggests polarization).
  - In a symmetric distribution: mean = median = mode.
  - In a skewed distribution: mode sits near the peak, not the center.
- Can identify the most common category or response.



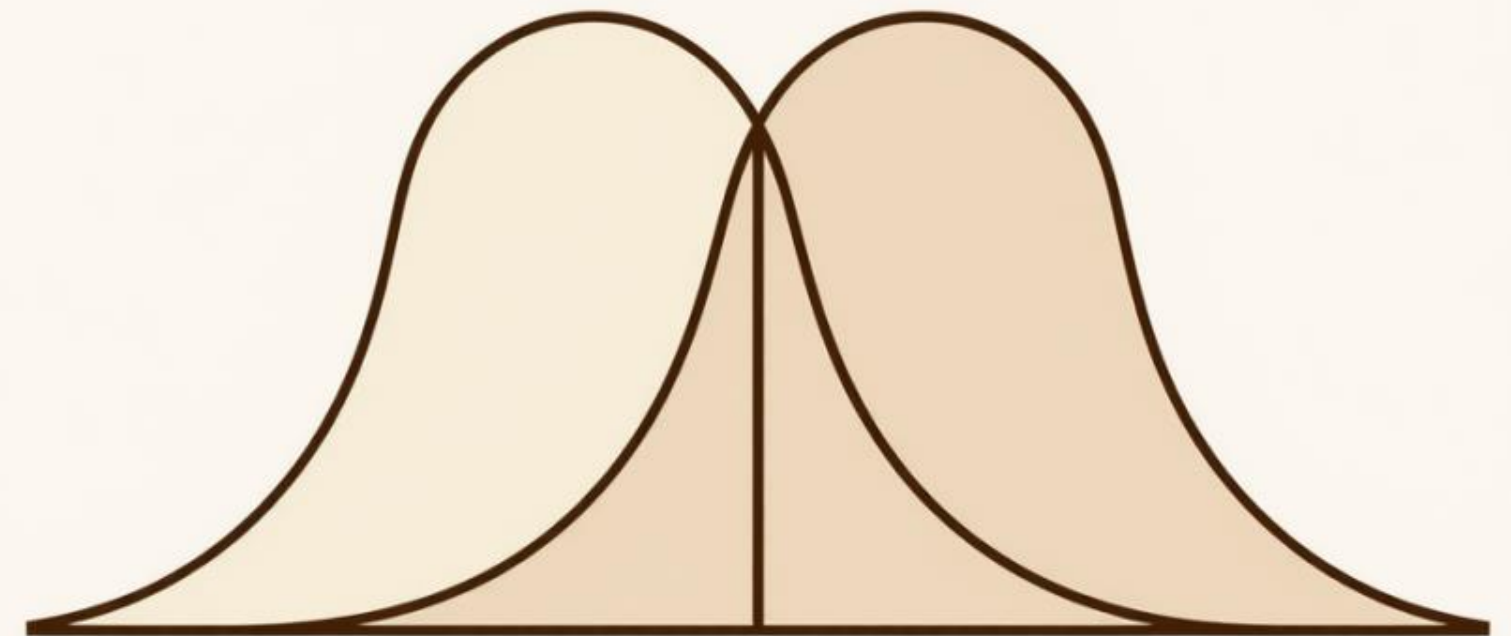
# Dispersion: Why It Matters

## Key Concept

- Central tendency alone is not enough to describe a distribution.
- Two variables can share the same mean yet look completely different in spread.

## Core Ideas

- Variability describes how far observations are from the center.
- Narrow distribution = low variability; observations cluster tightly.
- Wide distribution = high variability; observations spread far apart.
- Heavy tails indicate high variability and more extreme values.



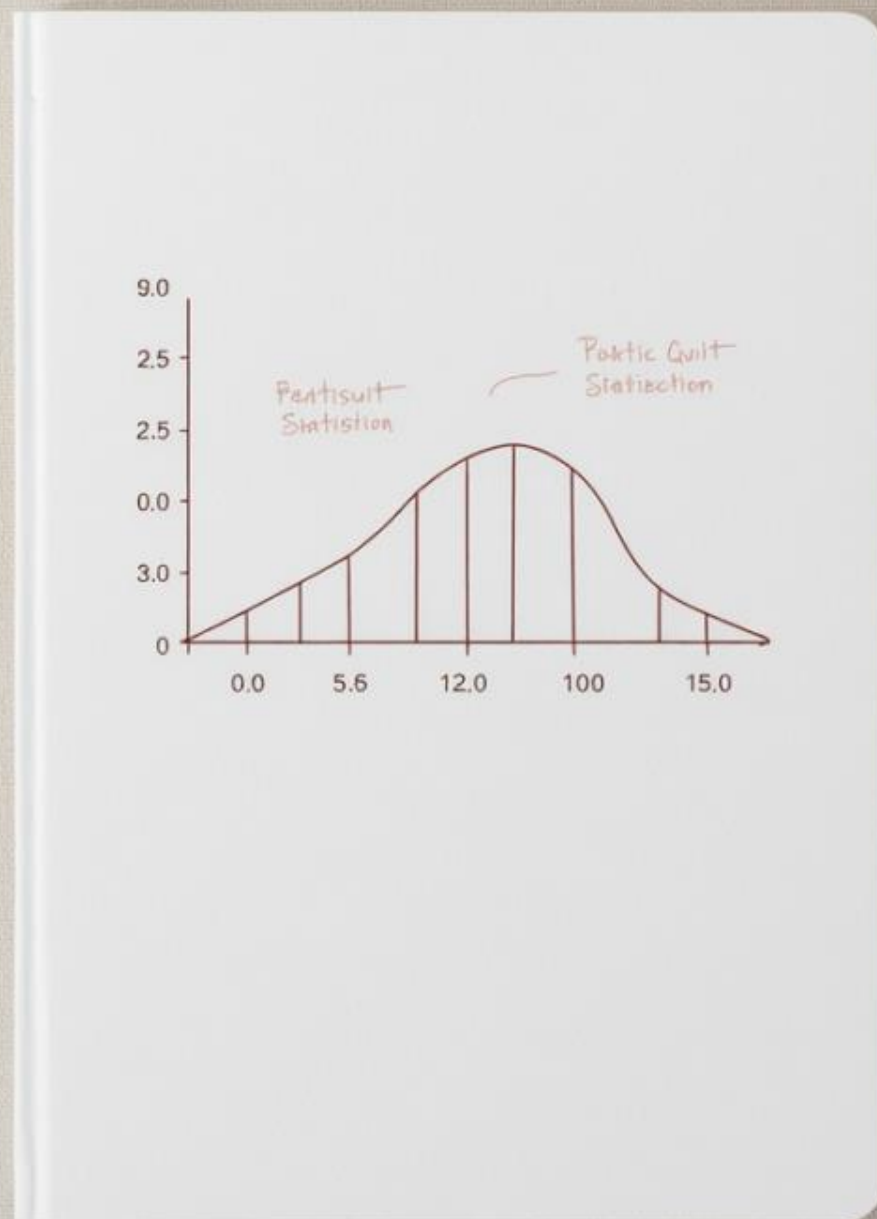
# Dispersion: Range and Standard Deviation

## Measures of Spread

- Range and standard deviation describe how far observations fall from the center.
- Range captures total spread.
- SD captures typical distance from the mean.

## Key Concepts

- Range = Max - Min; captures total spread, sensitive to extremes.
- Standard deviation (SD): average distance from the mean; weights large deviations more heavily.
- Larger SD = more spread; smaller SD = data clustered near mean, SD = 0 no variation.
- Both are sensitive to outliers and skewness.
- Empirical Rule (bell-shaped): ~68% within 1 SD, ~95% within 2 SD, ~99.7% within 3 SD



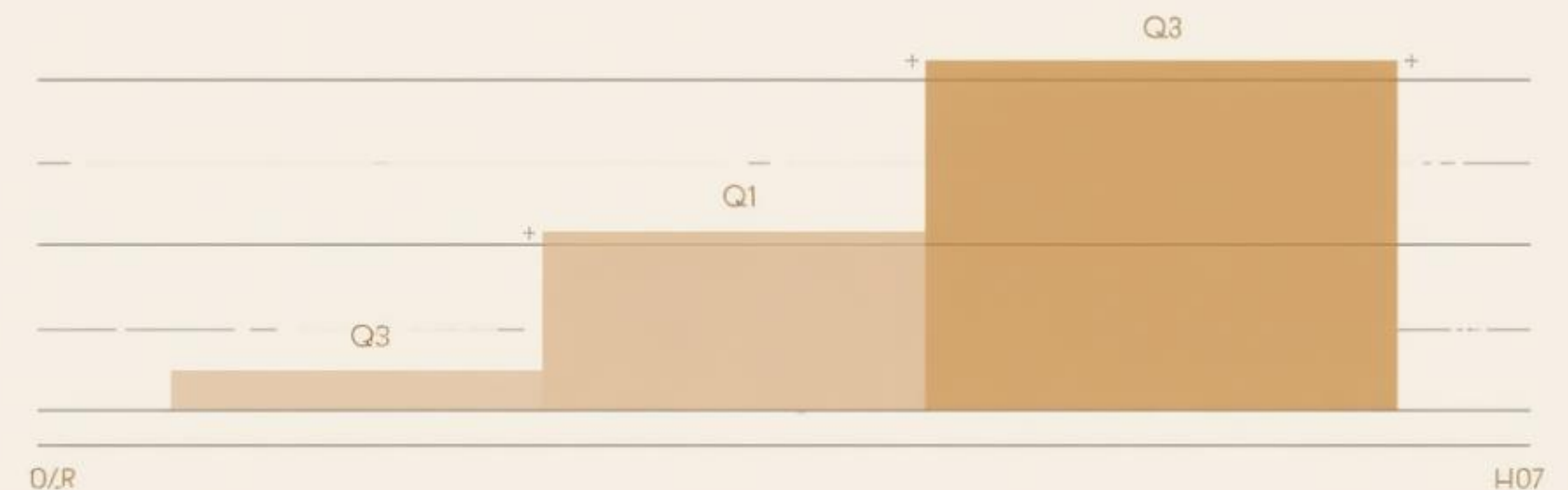
# Dispersion: Interquartile Range (IQR)

## Definition

- $IQR = Q3 - Q1$ , representing the spread of the middle 50% of data.
- Q3 is the 75th percentile and Q1 is the 25th percentile.
- It finds the distance between the 75th percentile (Q3) and the 25th percentile (Q1)

## Key Properties

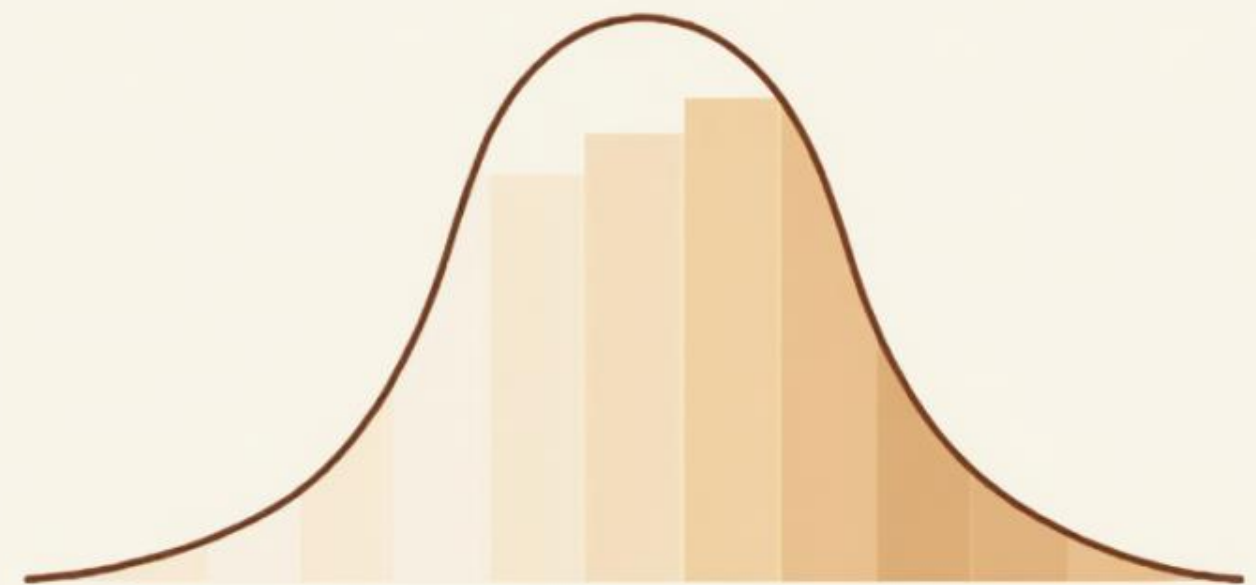
- Not affected by extreme values or outliers.
- Preferred measure of spread for skewed distributions.
- Larger IQR indicates more variability in the typical range.
- Boxplots visualize the median, quartiles, and IQR together



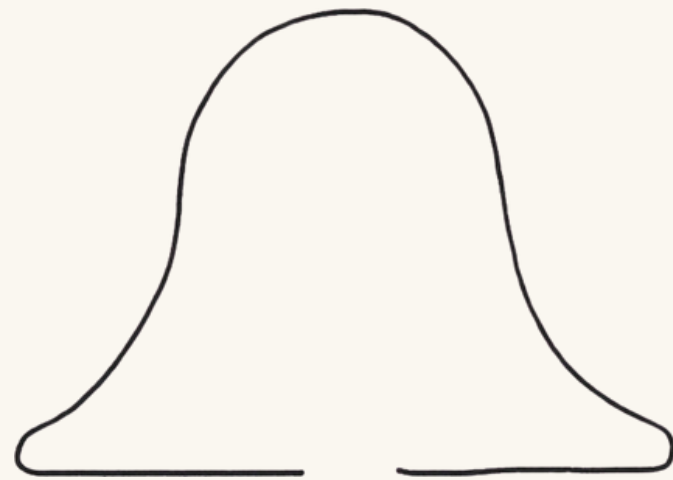
# Shape: What Is a Distribution?

- A distribution describes how values spread across observations.
- It can be shown using:
  - Tables (frequencies)
  - Histograms / density plots
- Key intuition:
  - Some values are common → “mass” of distribution
  - Some values are rare → “tails”

*Everything we compute (statistics like mean, median, SD) is a summary of this structure*

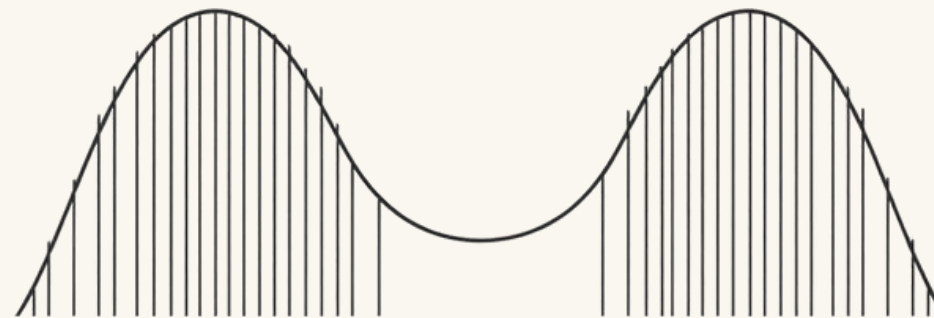


# Shape of a Distribution



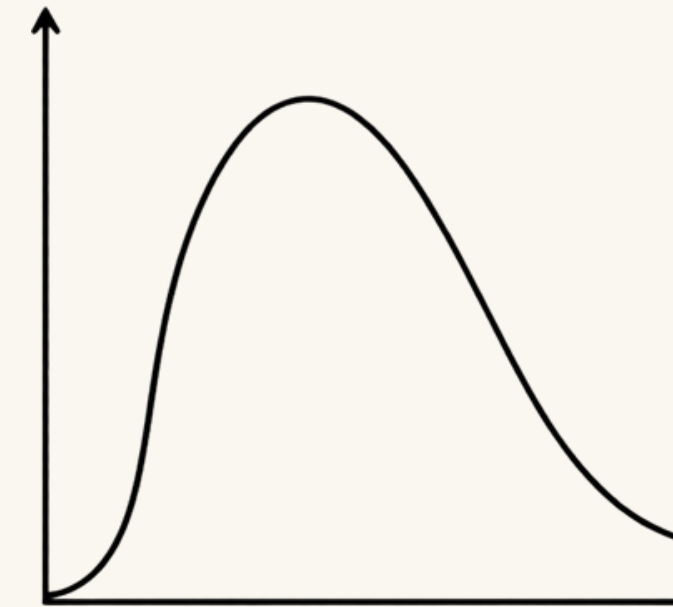
Bell-Shaped

- Symmetric distribution where most observations cluster around the center.
- Mean  $\approx$  Median  $\approx$  Mode. Common in test scores and natural phenomena.



U-Shaped

- High frequencies at both extremes with fewer observations in the middle.
- Often signals polarization or bimodal patterns in social data.

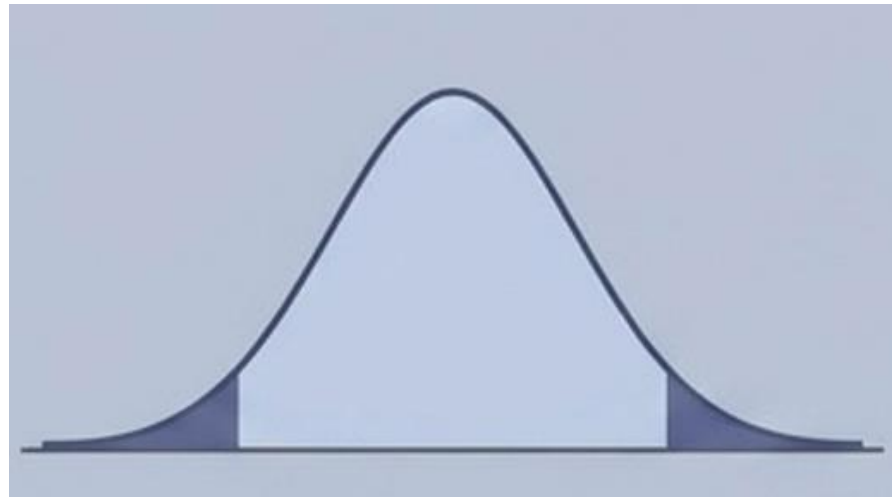


Skewed

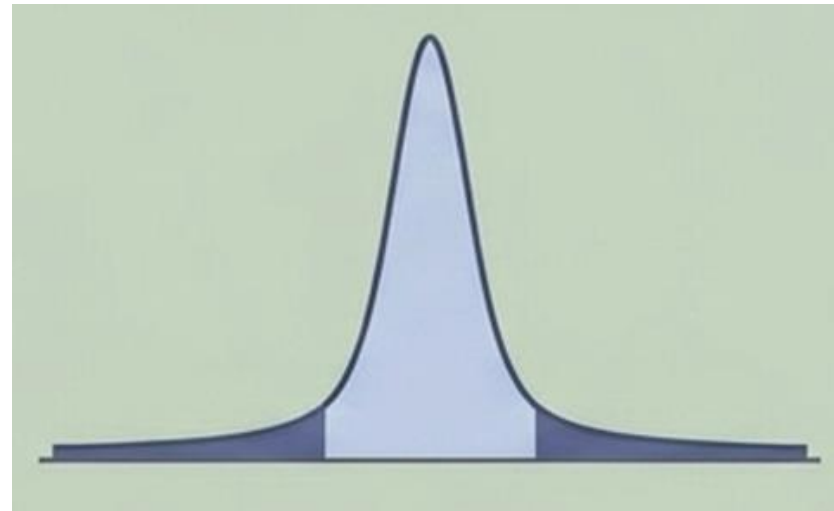
- Asymmetric tails pull the mean away from the median.
- Right-skewed: long right tail, mean  $>$  median.
- Left-skewed: long left tail, mean  $<$  median.

*Shape determines which summary statistics are most meaningful*

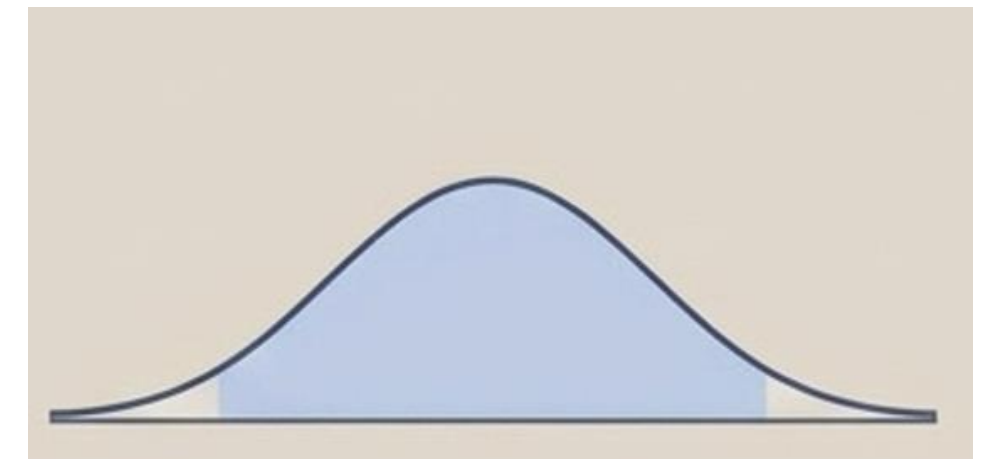
# Shape: Kurtosis



Normal (mesokurtic):  
moderate tails, baseline  
reference distribution.



High kurtosis (leptokurtic):  
heavy tails, more extreme  
values and outliers present.



Low kurtosis (platykurtic):  
light tails, fewer extreme  
values, flatter peak.

*Kurtosis describes the probability of extreme values — it tells us how heavy the tails are.*

# Putting It All Together

| Scale             | Type       | Display         | Central Tendency   | Dispersion                    | Shape              |
|-------------------|------------|-----------------|--------------------|-------------------------------|--------------------|
| Nominal           | Discrete   | Frequency table | Mode               | Variation in proportions      | N/A                |
| Ordinal           | Discrete   | Frequency table | Mode, median       | IQR, variation in proportions | “Skewness”         |
| Interval or ratio | Continuous | Histogram       | Mode, median, mean | Range, SD, IQR                | Skewness, Kurtosis |
|                   | Discrete   | Frequency table | Mode, median       | IQR, variation in proportions | “Skewness”         |

# Group Exercise

Work in groups of 4, discuss all three questions and be ready to share your reasoning. Using the table from the Geller & Curtis (2011), discuss:

**Table 1**  
Fathers' housing instability (year 5) by incarceration history.

|                                           | Full sample<br>N = 2763 | Ever incarcerated<br>N = 1052 | Never incarcerated<br>N = 1584 | Incarceration history unknown<br>N = 127 |
|-------------------------------------------|-------------------------|-------------------------------|--------------------------------|------------------------------------------|
| Any instability                           | 21%                     | 31%                           | 14%                            | 22%                                      |
| Instability index (0 = none, 6 = highest) | 0.29 [SD = 0.66]        | 0.45 [SD = 0.83]              | 0.18 [SD = 0.49]               | 0.30 [SD = 0.61]                         |
| Skipped Mortgage                          | 11%                     | 15%                           | 8%                             | 11%                                      |
| Moved in with others to save money        | 6%                      | 11%                           | 3%                             | 4%                                       |
| Moved more than once per year             | 5%                      | 8%                            | 2%                             | 8%                                       |
| Lived with others, not paying rent        | 3%                      | 5%                            | 2%                             | 2%                                       |
| Lived in shelter                          | 2%                      | 4%                            | 1%                             | 4%                                       |
| Evicted                                   | 2%                      | 3%                            | 1%                             | 2%                                       |

Analysis sample includes respondents answering all instability questions at year 5, excluding those incarcerated at year 5. All differences between formerly incarcerated and never incarcerated are statistically significant at  $P < .001$ .

1

What kind of variables are in the Geller & Curtis (2011) table?

2

Why is standard deviation reported for the Instability Index only? What makes mean & SD appropriate for that variable?

3

Interpret SD in context: what does a larger vs. smaller SD tell us about variation in the data?

# Next Class

- Bivariate descriptive statistics
- Basic statistical tests
- In-class quiz - 15 minutes

# Questions?

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15-minute break